What Master-Then-Reorganise Does Not Buy: An Invariance Argument for Body Codes and Its Two Boundaries

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ABSTRACT

Developmental robotics proposes mastering a competence then *reorganising* it for a new body. For a large family of interventions an identity settles it in advance. With the dynamics core frozen and only a low-dimensional body code θ adapted, replacing θ by Mθ for any invertible M and the code's weight block by $W_\theta M^{-1}$ leaves the function unchanged, so adaptation begun at $M\theta_0$ under an equivariant rule cannot change held-out error. We measure both conditions. **Linearity:** function-preserving *nonlinear* re-encodings cost +0.653 and +0.875 mm ($n = 20$, $p \leq 4 \times 10^{-4}$), while linear ones are bounded above by 0.020 mm, not resolved from zero. One is the identity at every training code, so it preserves leave-one-body-out decodability by construction, stays nonlinear where adaptation runs, and still costs +0.875 mm: probe and error come apart. **Optimiser equivariance:** under Adam even a rotation displaces error cell by cell. An ill-conditioned code's absolute cost grows with evaluation horizon on two disjoint seed blocks (+6.67 and +10.86 mm per log unit, $p \leq 2 \times 10^{-4}$); the relative cost is unresolved. Whitening the code costs +0.025 mm ($p = 0.68$) against a positive control resolving +1.234 mm. The code decodes named morphology scalars, though mass decodes only unstably. A fixed master-then-differentiate schedule is *worse* than joint training at 20 seeds on two of three bodies, under a post-hoc manipulation check. Our nulls are scoped to open-loop prediction on arm2. Whether it pays under a control objective is open; two attempts failed their pre-registered checks.

Index Terms — Developmental robotics, morphology transfer, negative results, reparameterisation invariance, world models.

I. INTRODUCTION

In developmental cognition it is widely held that agents progressively reformat implicit sensorimotor competence into more explicit, redeployable structure, and developmental robotics has pursued that idea for two decades [1], [2]. Karmiloff-Smith's *Representational Redescription* (RR) [3] frames development as a repeated cycle in which knowledge first embedded only in successful behaviour is later re-encoded into formats the system can access and recombine, placing that re-encoding *after* the skill has been mastered. The engineering proposal is direct: a model that has mastered one body's dynamics could be redescribed into an explicit, low-dimensional form, and transfer to a new body would be a matter of editing a few explicit variables instead of retraining.

We take an engineering operationalisation of that proposal and test it. RR proper is endogenous and triggered by the system's own mastery, whereas every mechanism we implement is experimenter-driven. Our results therefore bear on the engineering analogue, and the null below is neither evidence for nor against RR in humans (Section V).

The substrate is a deterministic, state-based world model of body dynamics: a delta-state predictor $f(s, a) = s + \Delta$, where s is proprioceptive state and a is muscle drive, trained on task-agnostic motor babble for a simulated muscle-driven arm with no pixels and no task reward inside the model. Bodies are dimension-preserving parametric perturbations of a base arm, scaling mass and inertia, joint damping and muscle-force gain, so state and action spaces never change and transfer is well-posed.

The mechanisms fall into two families and the asymmetry matters. Mechanisms 1 to 4 test *redescription*, the reformatting of an already-mastered model; mechanism 5 tests *developmental staging*, which is curriculum ordering. The two are developmentally distinct in Karmiloff-Smith's scheme, since RR reformats existing representations while a curriculum reorders experience, so we bundle them only as an engineering "master-then-X" family and do not treat a staging result as evidence about redescription-proper. The distinction is load-bearing: our one directional result is carried by mechanism 5, the looser analogue, while the one mechanism faithful to RR-proper that we ran at an interpretable size returned a null.

Terminology. By *body-code* (θ) we mean an experimenter-imposed, low-dimensional *input slot*, not an emergent hidden unit and not a variable the agent owns, optimised solely to absorb per-body dynamics variation. We recover only inertial and actuation scalars (mass, damping, gain) and no postural, geometric or kinematic content. This is not the spatial body schema of the neuroscience literature [4], [5], nor the robot body schema of Hoffmann et al.'s [6] review, which keeps a representation of where the body's parts are in space; ours is a body-*dynamics* parameterisation. *Emergent* means only that θ is never regressed onto morphology during training: we impose the low-dimensional compression, and only the code's *alignment* within that slot is emergent.

Contributions.

- **A corollary of two known results, applied to conditioning codes, with its magnitude measured.** Neither ingredient is ours: the GL(n) symmetry of a layer's input block is one of the parameter-space symmetries surveyed in [7], and the optimiser-equivariance taxonomy is textbook [8, §9.5.1], [9]. We add the corollary for a body-code, two conditions under which it fails (a moving core, or an under-determined fit), and the measurement that the same start-point difference is 0.0016 of a detectable effect under prediction but 2.14 of one under closed-loop control, a 1316-fold difference in what the two metrics can resolve

(Section IV-C4), which scopes our nulls to the prediction metric.

- **A construction that separates decodability from prediction accuracy.** A re-encoding that is the identity at every training code preserves leave-one-body-out decodability by construction (difference 0.000 on all twenty seeds) while remaining nonlinear where adaptation runs, and still costs +0.875 mm ([+0.474, +1.275], arm2). Decoding ground-truth factors is a standard proxy for representation quality, and [10] is a critique of the literature built on it; this is a counterexample to it on the metric we measured. Both halves come from the twenty-seed arm2 run of Section IV-C1, not a separate experiment.
- **A calibrated null, pooled over seeds and cells, on the second of the two routes the identity leaves open.** Those routes are a nonlinear map and conditioning under a non-equivariant optimiser. Against a positive control we bound any net benefit from conditioning, pooled over seeds and cells, at 0.099 mm, taken from the interval end most favourable to whitening (Section IV-C2). The control's excursion is larger than the repair whitening performs, the control map having condition number 300 against the whitening map's 5.2 at the median seed, so the bound rests on the measured interval and not on a matched-magnitude comparison.
- **A test of the master-then-differentiate axis on a learned dynamics world model** (a fixed schedule, not a competence gate), with a directional result. Staging is *worse* than joint training at 20 seeds on two of three bodies, against a joint arm cut to 1121 optimiser steps versus the staged arm's 1150, though the two are not equalised by learning-rate-weighted budget (Section IV-A, S8).
- **A factor-specific embodiment-dependence.** Damping and gain decode from θ on every body we measured; mass is unstable across seeds, reaching 0.805 on arm2 (12/12 seeds significant) and 0.58 on `myoArm` (4/4) against medians of −0.16 on arm3 (5/12) and −0.17 on arm4 (4/12). Even the arm3 and arm4 counts are far above the 0.6 of twelve expected by chance ($p = 1.8 \times 10^{-4}$, 2.2×10^{-3}), so what fails is reliability, not decodability. We do not claim monotone degradation with coupling.

Non-claims. We do not claim the first representational redescription in a neural network [11], nor that a decodable context code is novel (CaDM; RMA). We make no claim that the code is available *to the system itself*. We show only that morphology information is linearly recoverable by an external probe, which is strictly weaker. We do not claim self/world separation. The world model is body-only by construction.

II. RELATED WORK

Hidden-parameter and context-conditioned dynamics. A latent indexing a family of dynamics is the Hidden-Parameter MDP [12], and our body distribution is such a family. What matters is the distinction between parameter-fed and unsupervised codes. RMA [13] trains a latent whose *input* is the ground-truth privileged parameters, so decodability back to them is near-by-construction, and PrivilegedDreamer [14] conditions on explicit estimates of them. CaDM [15] infers a context latent from transitions and its latents separate visibly by ground-truth parameter under t-SNE, but it reports no quantitative recovery of *named* physical scalars; PEARL [16] amortises a per-task latent without recovering named parameters; Nagabandi et al. [17] meta-learn a fast-adapting model instead. Our characterisation sits in that gap: θ is fit unsupervised, never fed or regressed on morphology, yet named log-morphology scalars are linearly decodable from it post hoc. None of these tests a mastery or staging trigger.

Disentanglement and system identification. A deflationary reading is that θ decoding to log-mass, damping and gain is a disentanglement or system-identification result. Locatello et al. [10] prove unsupervised disentanglement *impossible* without inductive biases on both model and data, and classical system identification estimates physical parameters from a running estimator [18]. That literature offers a candidate account of mass's fragility, since under biarticular coupling a factor can become structurally non-identifiable, but our data do not test it: the most heavily coupled body we ran decodes mass best of the coupled bodies, at four seeds (Section IV-B).

Network-internal redescription, curricula and skill growth. Bartfai's [11] R2MAP implements agent-internal implicit-to-explicit reformatting in an ART network, and is why we do not claim the first redescription in a network; it is stronger than ours in being *triggered* by perceived task difficulty where our staged arm advances on a fixed schedule. DreamCoder [19] and Voyager [20] grow skill libraries from solved tasks, only Voyager gating on competence, and both add born-explicit skills instead of reorganising a learned world model. Staged-versus-joint training is an instance of curriculum-versus-i.i.d. training [21]. The automatic-curriculum literature runs from a typology of intrinsic motivation [22] to methods whose benefit is not uniform: some progress signals do worse than uniform sampling [23], and teacher algorithms differ in efficiency across learners [24]. Negative curriculum results are therefore not new; what is specific here is a fixed master-then-differentiate schedule applied to a learned dynamics model, with a mechanistic diagnosis.

Staged experience in development. The hypothesis is not ours. Elman [25] showed a recurrent network could learn a grammar it otherwise failed on when its effective memory was grown on a fixed schedule; Rohde and Plaut [26] re-ran that setting and the advantage did not survive. Bongard [27] moved the idea from input to body, and his is both the closest precedent to our question and the one clear positive: simulated robots grow from an anguilliform to an upright body plan within a single evaluation period, on a developmental window the experimenter shortens across four successive evolutionary stages, and populations staged this way evolve phototaxis gaits faster and lose less distance under random perturbation. That schedule is fixed, as ours is, and not under selection. Two differences explain why the positive does not predict ours. His schedule acts on an evolutionary search over controllers, where the path a population takes changes what it can reach; ours reorders the data a single fitted model sees, where the objective and its optimum are unchanged by the ordering. His currency is robustness to perturbation, not prediction accuracy on a held-out body, so a staging benefit in his setting is consistent with a staging cost in ours. Clark and Karmiloff-Smith [28] supply the representational half, in which the currency is accessibility, not task success. Two consequences follow for our design. All three of these proposals specify an ordering of experience on a fixed schedule rather than a competence gate, which is why our staged arm advances unconditionally and why we make no mastery-trigger claim; and their currency is accessibility while our dependent variable is transfer accuracy, a mismatch we do not resolve and treat as a scope limit (Section V).

Parameter-space symmetry and optimiser equivariance. Two facts we use are established and we claim neither. A layer's input block carries a GL(n) symmetry, surveyed with the other parameter-space symmetries by Zhao et al. [7], and Neyshabur et al. [29] built Path-SGD on a different symmetry but the same underlying point, that gradient descent is not invariant to function-preserving reparameterisations. Whether the symmetry carries from the function to the *estimate* is governed by the optimiser. Newton's method is affine-invariant [8, §9.5.1]; BFGS is not in general, but on this problem we measure L-BFGS to be effectively invariant (Section IV-C1); Adam is equivariant only under permutations because its second moment is an elementwise square, and we refer to Ling et al. [9] for the rotation failure and to Section IV-C1 for the rescaling failure, which we verify numerically ourselves. Our contribution is neither fact but their corollary for a conditioning code, with the magnitude measured on this substrate.

Adjacent lines. Robot body-schema and visual self-modelling systems learn explicit kinematic or dynamic self-models via sensorimotor calibration ([6], [30], [31]); these represent a self/non-self boundary and ours does not, the world model being body-only by construction. Controllable-latent world models factorise visual dynamics into controllable and non-controllable parts [32], [33], with no mastery-triggered reformatting, and symbolic-distillation methods extract closed forms from learned dynamics [34], [35]. Those two lines are both experimenter-run, and neither has a mastery trigger. Three of our five mechanisms are distillation-based, and the one with reportable data is a null whose interval excludes a redescription advantage larger than about 0.25 mm (Section IV-A). Finally, fine-tuning has very low intrinsic dimension [36] and LoRA [37] exploits this by confining the update to a low-rank subspace; a per-body code over a frozen shared core is structurally the same object, which is why the interventions we test have little room left to operate (Section V).

III. METHOD

Bodies. Planar muscle-driven arms in MuJoCo with 2, 3 and 4 links (arm2, arm3, arm4; state dimensions 8, 14, 18) plus MyoSuite's `myoArm` (20 joints, 34 muscles, 74-dimensional state) and `myoHand` (23 joints, 39 muscles). Perturbations scale link mass and inertia, joint damping, and muscle-force gain, leaving state and action dimensions unchanged so that transfer is well-posed.

World model. A delta-state MLP $f(s, a; \theta) = s + g([\mathrm{norm}(s), a, \theta])$, where g is two hidden layers of 128 SiLU units with input and output normalisers fitted on training data. Training data is task-agnostic motor babble under an Ornstein–Uhlenbeck action process (mean reversion 0.15, noise scale 0.4, actions clipped to $[-1, 1]$, pose re-randomised every 100 steps), 6000 transitions on each of 10 training bodies. The rounds reported here train the core for 25 epochs of Adam at 10^{-3} with batch size 1024. There is no reward and no pixel input anywhere in the model.

Body-code model. A single shared core g is trained jointly over a set of training bodies, each carrying its own free code $\theta \in \mathbb{R}^4$, initialised at $0.1 \times N(0, I)$ and optimised only to reduce prediction error, never regressed onto morphology. The training bodies are the nominal one plus nine log-uniform draws scaling mass by [0.6, 1.7], damping by [0.4, 3.0] and gain by [0.6, 1.6]. At test time the core is frozen and only θ is adapted to a held-out body from a small budget of transitions.

The five mechanisms. (1) LPV distillation: redescribe the mastered MLP into a local linear-parameter-varying model, versus fitting an LPV model directly to the same data. (2) Broad-support distillation: as (1) but over a wider state support generated by the MLP itself, fixing the same-support confound. (3) Few-shot explicit handles: adapt only 2·(state-dim) physical knobs instead of full fine-tuning. (4) Intervention-based distillation: fit the bottleneck by querying the frozen mastered model at chosen inputs, versus fitting the same bottleneck on the model's natural nominal data. (5) Developmental staging versus joint: a staged learner masters the nominal body then progressively differentiates to new bodies. A joint learner trains on all bodies at once. Few-shot adaptation then adjusts only θ to held-out bodies. Stage advance is an unconditional loop over a fixed epoch count. There is no competence gate anywhere in the code, so we make no mastery-trigger claim. Only mechanisms 4 and 5 carry reportable data; Section IV-A gives the record for the other three and withdraws their numbers.

Metrics. The primary transfer metric is open-loop fingertip prediction error at horizon $k = 6$ on a held-out body after adapting θ, in millimetres, and where a budget sweep is involved an area under the adaptation curve. We additionally report a random-shooting MPC metric and learned-policy SAC+Dyna RL return.

Statistics and pre-registration. Minimum detectable effects are computed as $2.93 \times \mathrm{SD}/\sqrt{n}$ at the round's own n, where $2.93 = t(.975, 19) + z(.80)$ fixes the 80% criterion at the project's design size; projected and achieved values are kept distinct wherever either is quoted. The unit of analysis is the seed. *A cell that fails its own manipulation check is not interpreted*, including when its unused numbers would have supported us, and Section IV-A and Section IV-C each contain an instance. *Not every result below sits under a pre-*

registration, and we name those that do not. The five-seed staged-versus-joint comparison in Section IV-A, the staged-versus-joint figures closing Section IV-B, both of Section IV-D's control metrics and Section IV-B's twelve-seed decodability result are exploratory and marked as such where reported. Four of the ten settings in the difficulty round are not covered by a pre-registration; their design document postdates their results. The measurements stand. The guarantee does not extend to them. S6 gives the conventions and the coverage in full.

Re-encodings of the code. A re-encoding is applied by a wrapper that stores θ in the mapped space and applies the map's inverse before handing the code to the frozen core, so for corresponding codes the wrapped model computes the same function of state and action; Section IV-C1 verifies this across all five re-encodings to 3.8×10^{-6}. Five re-encodings are used, all at $\theta_{\text{dim}} = 4$; S3 gives the specification in full.

`organised` is the identity. `k1` and `k300` are linear, $\theta \mapsto \theta A^{T}$ with log-spaced singular values of geometric mean 1 and condition number $\kappa \in \{1, 300\}$. `scrambled` and `nonlinear_readable` are four-layer additive couplings of the RealNVP form and share their weights with each other. `nonlinear_readable`, the *gated* map, multiplies each layer's displacement by a factor that is exactly zero at every training code, so the composition is the identity there and leave-one-body-out R^2 is preserved by construction, while the map stays nonlinear where adaptation runs. The gate width is a tuned hyper-parameter and was not pre-registered: the selection rule is monotone by construction and could only ever have returned the smallest candidate, so Section IV-C1 reports the cost at all three calibrated widths. S3 gives every map in full.

The conditioning of a nonlinear map is not a free parameter, so we measure it rather than assume it. The median condition number of the Jacobian, over 24 probe codes per seed, is 90.9 for the gated map (34.2–285.1 across seeds) and 37.7 for `scrambled` (12.0–279.1), against 300 by construction for the linear `k300` control. The linear control is therefore the *worst* conditioned of the three, which is what makes the comparison interpretable. Whatever the nonlinear maps cost, conditioning is not the explanation.

Adaptation and its start. Few-shot adaptation fits θ only, minimising one-step prediction error on the normalised state delta with no penalty on θ; the reported metric is a different quantity, open-loop fingertip error after six steps. Two update rules are used, L-BFGS and Adam at three learning rates. Every arm begins from $M\theta_0$, the image of the same point under that arm's own re-encoding, so all arms start from the *same function*. S5 gives both optimisers' settings. The comparison therefore isolates the geometry the optimiser works in and nothing else, which is what makes the L-BFGS column of Fig. 1 a test of the identity, not of the start. Under Adam the same construction still fails to be free, and that is the point. Satisfying the identity's precondition is not sufficient when the adaptation rule is not equivariant under the map.

IV. RESULTS

A. *Staged versus joint training*

Two of the five mechanisms carry reportable data. Mechanisms 1–3 were run at two seeds with no stored result file, and the closest stored JSON disagrees with the surviving lab note. Given that sample size and provenance, the numbers support no claim, and we do not report them. We retain from those three only the *design* contribution. Each fixes a specific confound that motivated mechanism 4, which was then run properly. Readers should treat the five-mechanism framing as two mechanisms with reportable data and three design precursors.

Mechanism 4, intervention-based distillation, is a CI-bearing null. The stored quantity is `direct − redescription`, whose value is −0.224 mm with a 30-cell bootstrap of [−0.750, +0.319] (`results/v2powered_arm2.json`). We report it in the opposite orientation, so that a positive number means redescription is worse. Redescription is therefore 0.224 mm *worse*, seed-level CI [−0.245, +0.693] ($n = 5$, $p = 0.26$), straddling zero and excluding a redescription advantage larger than about 0.25 mm. The 30-cell bootstrap in that orientation is [−0.319, +0.750], at the wrong level. Its 30 cells are 6 fixed single-factor perturbations crossed with 5 seeds, not 30 bodies sampled from the distribution as in mechanism 5, so its effective body diversity is 6. The interval resamples those clustered cells, not the seeds. Its redescription arm receives *extra* query signal, so the null is conservative.

Mechanism 5, staging versus joint training, is directional, and staging is worse. On the best-scaled comparison (arm2, 10 training bodies, 5 seeds, 30 held-out bodies; exploratory, Section III), joint held-out adaptation AUC is 4.44 mm against staged 5.34 mm, a difference of −0.90 mm. At the seed level the interval is [−1.593, −0.203] ($n = 5$, $p = 0.023$). A 30-cell bootstrap would give [−1.27, −0.55], 1.9× narrower and at the wrong level. Staged wins on only 20% (6/30) of held-out bodies. The marginal per-cell distributions overlap almost entirely (joint [1.76, 7.09] against staged [2.12, 8.97]).

The staged and joint arms do not receive equal optimiser steps. The staged arm takes 78.0% of the joint arm's, or 30.5% once each step is weighted by its learning rate. The gap between the two accountings is the 0.3× core rate at which the differentiation stages run. That reduced rate *is* the curriculum's operational definition of consolidate-then-differentiate, so equalising the weighted budget would abolish the manipulation; what can be done is to raise it and see whether the direction turns. `DESIGN_MATCH.md` fixes five training conditions in advance, and the direction survives on both bodies at every budget while the magnitude attenuates on arm3, from −1.238 mm at the 40% weighted budget to −0.811 mm at 69%, a 34% change across the tested range. At the 40% weighted budget, seventeen of twenty seeds favour joint training on arm2 and eighteen of twenty on arm3. The trend does not reach zero inside the tested range (S8), so the residual step disparity does not plausibly account for the direction, but we cannot say the effect is independent of it. S8 gives the five conditions, the per-budget table and the 4-seed arm3 run.

A third body is excluded, by a criterion that is not the one we pre-registered, and its excluded numbers would have supported us. The pre-registration required θ-adaptation to beat no-adaptation *on every seed*. That clause is defective. It gets harder as n grows, and at $n = 20$ the best arm on arm2 reaches only 19/20, so applied literally it voids all three bodies, including the two that carry our directional result. We therefore substituted a criterion, the seed-level 95% interval for (adapted − unadapted) lying wholly below zero, and applied it uniformly to all three bodies × five training conditions. Under it, arm2 and arm3 pass on every arm (mean improvements 1.1–3.0 mm, $p \leq 2 \times 10^{-3}$) and arm4 fails on every arm (five contrasts straddling zero, $p = 0.14$–0.27), so no arm4 cell may be interpreted. Two disclosures follow and neither is comfortable. The substitution was made after the per-seed clause was seen to fail at $n = 20$, and the substitute criterion was fixed after the per-body outcomes were visible. It costs us arm4, whose direction agreed with ours, but a reader is entitled to weigh that we changed a pre-registered rule and that the change is what keeps our own result alive.

The two families are asymmetric in a way worth stating. Mechanisms 1–4 are the *closer* analogue to representational-redescription-proper: the one with reportable data is a CI-bearing null, and the other three were not run at a size that supports any statement. The directional worse-than-joint result is carried by mechanism 5, which is *curriculum staging*, the looser Karmiloff-Smith analogue.

B. *The morphology-aligned body-code*

Training a shared core jointly across bodies with a per-body code (θ, dimension 4) yields a code from which named log-morphology scalars are linearly decodable under leave-one-body-out cross-validation. On arm2, over $n = 12$ seeds with no exclusions and a 200-permutation null for each (exploratory, Section III): log mass $R^2 = 0.805$ [0.091, 0.947] (12/12 seeds $p < 0.05$), log damping 0.916 [0.312, 0.976] (12/12), log gain 0.910 [0.833, 0.969] (12/12). Brackets are full per-seed ranges, not intervals.

The non-obvious element is *which factor breaks* (Fig. 2), measured across three embodiments at matched dataset size (20 bodies × 12 seeds each). Damping and gain decode strongly on all three (medians 0.92/0.97/0.98 and 0.91/0.94/0.86; 12/12 seeds significant on every body and factor). Mass is unstable across seeds: arm2 median 0.805 with 12/12 seeds significant, against arm3 −0.16 (5/12) and arm4 −0.17 (4/12), with individual seeds spanning −0.79 to +0.73 across the two bodies (arm3 [−0.74, +0.73], arm4 [−0.79, +0.69]).

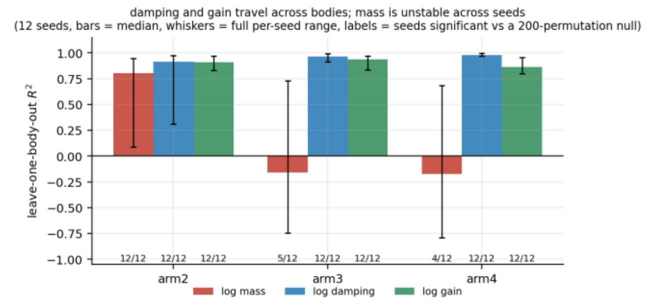

Fig. 2. Leave-one-body-out decodability of named morphology scalars from the unsupervised code, three bodies, 12 seeds. Damping and gain travel everywhere; mass decodes on arm2 but is unstable across seeds on arm3 and arm4, where the median is negative and 5/12 and 4/12 seeds still clear their own null. Labels give the number of seeds significant against a 200-permutation null.

The pattern is not monotone in mechanical coupling. `myoArm`, whose 34 muscles actuating 20 joints make it by far the most coupled body we ran, decodes mass at 0.58 with 4/4 seeds significant, better than either planar coupled arm. What survives is a factor-specific description, weaker than any claim of a trend. Damping and gain are robustly decodable everywhere we looked; mass is not *reliable*. On arm3 and arm4 its median is negative, yet 5/12 and 4/12 seeds clear their own permutation null, far above the 0.6 of twelve expected by chance ($p = 1.8 \times 10^{-4}$ and 2.2×10^{-3}), so the supported description is instability across seeds, not failure to decode. A label-shuffle null is negative on all three bodies, and a $\theta_{\text{dim}} = 3$ ablation leaves the arm2 figures essentially unchanged (0.77/0.91/0.90), so the alignment is neither a cross-validation artifact nor an effect of a slack code dimension.

On arm2, joint training aligns better than staged on all three factors (staged 0.46/0.63/0.77 against joint 0.82/0.91/0.90; exploratory, Section III). This does not hold on the other two bodies: on arm3 staged decodes mass better (0.38 against 0.15), and on arm4 staged is ahead on gain (0.87 against 0.85). The narrow finding that survives is that on arm2 the jointly trained code aligns better, in exploratory runs at unmatched optimiser budget. That contrast does not separate how much body variation the code saw from the order in which it arrived, and it reverses on the other two bodies. All six of those figures come from 3–4-seed runs, below the 12 seeds used above.

C. *Re-encoding invariance and its boundaries*

Fix the shared core and adapt only the code, as mechanisms 4 and 5 do and as any method that fits a free context code by gradient descent does. For any invertible M, set $\theta \mapsto M\theta$ and $W_\theta \mapsto W_\theta M^{-1}$, where W_θ is the single weight block the code enters. Then $f(s, a; \cdot)$ is unchanged for every state, action and body. Adaptation from the mapped start $M\theta_0$ therefore minimises the same objective on the same data, and if its update rule is equivariant under M the estimate becomes $M\hat{\theta}$, with $f(M\hat{\theta})$ bit-identical to $f(\hat{\theta})$. Under those two conditions held-out error cannot change. A further two conditions hold in our setup and are stated so that a reader can check them elsewhere. The adaptation objective carries no penalty on θ (an L2 penalty is not invariant under M), and θ does not pass through the input normaliser, which is applied to the state alone. Any intervention that re-coordinatises the code by an invertible matrix, whether by orthogonalising, rescaling or whitening, is inert by construction in this regime. Nonlinear re-encodings fall outside the family, and disentangling generally requires one. We measure one below at +0.875 mm.

1) The identity and its measurement:

Over 20 seeds and three adaptation budgets under L-BFGS, a random orthogonal reparameterisation costs −0.00007 mm (seed-level CI [−0.00015, +0.00001], $p = 0.075$) and an affine map with condition number 300 costs +0.006 mm ([−0.008, +0.020], $p = 0.36$). Both numbers should be read against the construction's own numerical floor. The five re-encodings agree as functions to 3.8×10^{-6} in the units of the predicted state vector (joint angles in radians, joint velocities and muscle activations), which, propagated through this body's kinematics, is worth about 1.4×10^{-3} mm of fingertip displacement (median over 200 randomly drawn poses). The rotation map's point estimate is twenty times *smaller* than that. This bound is on how far the arms' functions can differ, not on how precisely a difference can be measured. A quantity known in advance to be tiny can still carry a narrow interval, which is why the rotation map's is an order of magnitude narrower than the bound itself. What the bound does establish is that a systematic displacement of 7×10^{-5} mm is fully accounted for by the arms not being exactly bit-identical, so we read it as a bound rather than as a cost and its $p = 0.075$ does not bear on the identity. We apply that consistently. The 0.00017 mm we quote for the rotation under L-BFGS below, and the +0.0001 mm for whitening under L-BFGS in Section IV-C2, sit beneath the same bound and are likewise consistent with exact inertness, not with evidence of a small real effect. The `k300` arm at +0.006 mm is about four times above the bound and the two nonlinear maps two to three orders of magnitude above it, so the separation this section rests on is between quantities the construction distinguishes.

The two function-preserving but nonlinear re-encodings cost +0.875 mm ([+0.474, +1.275], $p = 2 \times 10^{-4}$) and +0.653 mm ([+0.333, +0.973], $p = 4 \times 10^{-4}$). Linear re-encoding is thus two to three orders of magnitude cheaper than nonlinear (Fig. 1), even though every one of these maps leaves the function unchanged (all five re-encodings agree to 3.8×10^{-6}). The identity's linearity condition does real work. The gated map sharpens this. It preserves the code's leave-one-body-out decodability *exactly* (difference 0.000 on all 20 seeds) and still costs +0.875 mm, so decodability is not the operative variable.

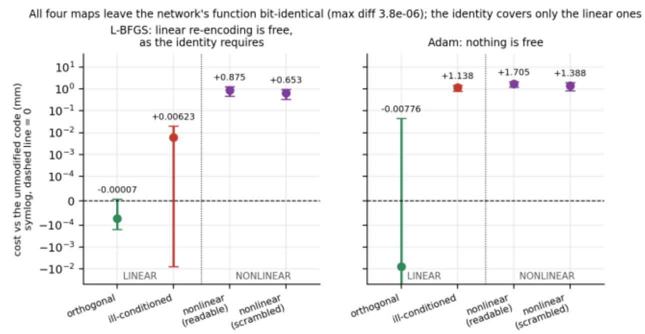

Fig. 1. Linear against nonlinear re-encoding, arm2, 20 seeds, three adaptation budgets pooled, log scale. All four maps leave the network's function unchanged, so what separates them is linearity and not function preservation. Under L-BFGS the linear maps sit at or within four times the construction's numerical floor while the nonlinear maps sit two to three orders of magnitude above it; under Adam even the linear maps stop being free.

Under Adam nothing is free, and that bounds what the identity buys in practice. The ill-conditioned map costs +1.04 to +1.22 mm ($p < 10^{-4}$), the same order as the nonlinear re-encodings. That range is across three learning rates (0.015, 0.05 and 0.15) swept with no pre-registered choice among them; 0.05 is the middle value on every arm in this round (`k300` gives +1.225 / +1.138 / +1.037 at 0.015 / 0.05 / 0.15), so the sweep is an unregistered degree of freedom we did not exercise in our own favour. The most favourable rate for the nonlinear result is 0.15, where the gated-minus-scrambled contrast reaches +1.272 mm ($p = 0.001$) against +0.317 mm ($p = 0.20$) at the rate we report. L-BFGS takes no learning rate, so the figures above are unaffected. The orthogonal map's *signed* seed-level mean is near zero ($p \geq 0.195$), but that is cancellation, not invariance: cell by cell it displaces held-out error by 0.036–0.512 mm on average across nine (learning-rate, budget) cells and up to 4.28 mm in a single cell, against 0.00017 mm under L-BFGS.

Adam's per-coordinate second moment is an elementwise square, which does not commute with a rotation, so Adam is equivariant only under permutations. The rotation failure is Ling et al.'s [9]. The rescaling failure is our own measurement. Plain gradient descent and momentum are orthogonally equivariant first-order methods, which follows from their updates being linear in the gradient and which we check numerically rather than take on trust from the citation. Over 200 random four-parameter least-squares problems, we mapped the estimate back and measured $\|M^{-1}\hat{\theta}(M) - \hat{\theta}(I)\|$. Adam's median displacement is 5.0×10^{-2} under an orthogonal map and 7.1×10^{-2} under a diagonal rescaling, against 2.7×10^{-16} for momentum under the orthogonal map. Both rules are zero to machine precision under a permutation (`src/verify_equivariance.py`, `results/equivariance.json`).

This is a property of the adaptation *path* at a finite budget, not of the optimum. On this convex problem every displacement falls to machine precision by 1000 steps, because the least-squares solution does not depend on the parameterisation. What distinguishes the rules at the 80 steps our mechanisms use is that momentum's orthogonal displacement is machine zero at *every* budget, because its update is linear in the gradient and commutes exactly with an orthogonal map, while Adam's second moment does not (`results/equivariance_budget.json`). The honest scope is therefore: *re-encodings the adaptation rule treats equivariantly are provably free; others are not, and conditioning becomes a real channel.*

Gate width. We re-ran the whole experiment at the two widths the selection rule could not have chosen. The gated map costs +0.875, +0.878 and +0.733 mm at 0.12, 0.18 and 0.25 ($n = 20$, $p \leq 6.5 \times 10^{-4}$), against the ungated +0.653 mm in all three runs. No pair of widths is distinguishable, and the comparison excludes differences larger than about 0.4 mm (S4).

2) Conditioning, against a positive control:

Two routes could still pay: a nonlinear map, which the identity does not cover and which costs +0.875 mm even under L-BFGS, and conditioning under a non-equivariant optimiser. We bounded the second, pooled over seeds and cells; a benefit concentrated on the worst-conditioned seeds is not excluded (S7). With the core frozen we whitened the code, $\theta \mapsto M\theta$ with $M = \Lambda^{(-1/2)}U^{T}$ built from the covariance of the training codes only (using the held-out body would be reading the test set), against three controls on identical seeds, bodies, data, starts and step counts: identity, a random orthogonal map, and an affine map with condition number 300. The round was pre-registered: arm2, 20 seeds × 4 held-out bodies, budgets {25, 100, 400}, under Adam (lr = 0.05) and L-BFGS. All four manipulation checks passed: the arms agree to 3.1×10^{-6}, the orthogonal map is null under both optimisers (+0.022 mm, $p = 0.38$ under Adam; −0.00004 mm, $p = 0.13$ under L-BFGS), and the ill-conditioned map is null under L-BFGS (−0.0055 mm, $p = 0.66$) as the identity requires. The fourth is not an ordinary check but a positive control. On the same seeds and budgets the ill-conditioned map reproduces the known damage under Adam at +1.234 mm, $p = 2.2 \times 10^{-6}$, demonstrating that the instrument resolves a conditioning effect of about 1.2 mm.

Against that instrument, whitening buys nothing (Fig. 3). At the seed level, with the three budgets pooled, `WHITE − ORIG` is +0.0246 mm under Adam (CI [−0.0986, +0.1479], $p = 0.68$), of the wrong sign and bounded well below what the instrument can see. Under L-BFGS it is +0.0001 mm ($p = 0.61$). A pooled mean of that kind can sit near zero through cancellation, as Section IV-C1 shows for the rotation, so we report this arm cell by cell as well. Under Adam its mean absolute displacement is 0.409, 0.359 and 0.225 mm at budgets 25, 100 and 400, with a largest single seed-and-body displacement of 3.05 mm, so the pooled mean is concealing movement of the same order as the rotation's. What it is not concealing is a benefit. The signed per-budget means are −0.037, +0.055 and +0.056 mm, so no budget shows a systematic gain, whereas the ill-conditioned control displaces 1.410 to 1.627 mm per cell with every budget signed the same way (+1.163 to +1.325). Under L-BFGS the whitened arm moves by at most 0.001 mm, as the identity requires. The bound that follows is therefore on the net effect pooled over cells, not on the magnitude of per-cell movement. Mean held-out error under Adam is `ORIG` 3.823, `WHITE` 3.848, `ILL` 5.057, `ROT` 3.845 mm. This is not a case of there being nothing to fix: the condition number of the training-code covariance has median 27.1, ranging from 4.1 to 243.7 across seeds. The whitening map that removes it therefore has condition number 5.2 at the median seed and 15.6 at the worst, against 300 for the ill-conditioned control, so the control is calibrated at a larger excursion than the repair performs.

Whether a benefit appears on the badly-conditioned seeds and is diluted by pooling is a question we did not pre-register. We answer it anyway and report the answer whatever it is: the seed-level slope on log condition number is **−0.063 mm per log unit** (95% CI [−0.170, +0.044], $p = 0.23$), the sign the dilution hypothesis predicts but not resolved, and flat under L-BFGS as the identity requires. This analysis is post-hoc and we label it so (S7, `src/whitening_moderator.py`).

This is therefore not "we did not find it" but a bound: on an instrument that resolves a +1.2 mm conditioning effect, moving the code into its own natural coordinates is worth +0.02 mm. Our earlier searches carried no positive control, so their nulls could not separate "no effect" from "an effect this measurement cannot see." This one can. We offer this as an observation rather than a claim, and only as an order of magnitude, since the two excursions are not matched: the point estimates differ about fiftyfold and the interval bounds any repair at one-twelfth of the damage. That bound is on the pooled contrast. The post-hoc moderator above neither establishes nor excludes a larger benefit on the worst-conditioned seeds, and its point estimate extrapolated across the observed range of that condition number would exceed it. Establishing that reparameterisation damage is asymmetric would need a design that asks the question up front.

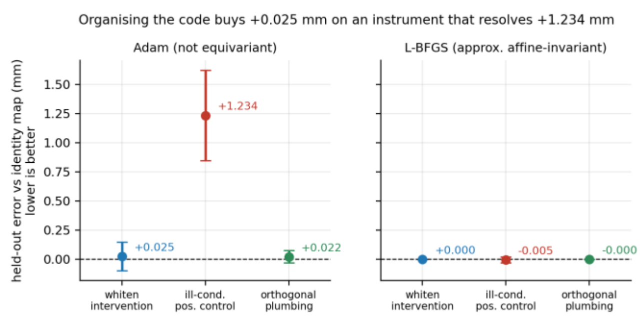

Fig. 3. The conditioning route, tested against a positive control. This is a different experiment from Fig. 1 (20 seeds × 4 held-out bodies, separately pre-registered), so its ill-conditioned map need not match Section IV-C1's +0.006 mm. Whitening the code is worth +0.025 mm ($p = 0.68$) on an instrument that the ill-conditioning control shows can resolve +1.234 mm ($p = 2.2 \times 10^{-6}$). Under L-BFGS every arm is null, as the identity requires. Bars are seed-level 95% intervals, $n = 20$.

3) Horizon scaling and cross-body scope:

The absolute cost of an ill-conditioned code grows with the evaluation horizon, and the measurement repeats on a second, disjoint seed block. With the adaptation held fixed and only the evaluation horizon varied ($k \in \{6, 20, 50\}$, arm2; Fig. 4), the ill-conditioned map costs +1.24, +7.17 and +15.58 mm, a seed-level slope against log k of +6.67 (CI [+3.62, +9.73], p =

2×10^{-4}). On twenty fresh, disjoint seeds the same measurement gives +1.91, +14.18 and +25.01 mm, slope +10.86 (CI [+6.24, +15.47], $p = 1 \times 10^{-4}$), a larger effect than the first block's. Neither block resolves whether the *relative* cost also grows. As a fraction of baseline the cost is 0.370/0.338/0.308 on the first block (slope −0.029, $p = 0.41$) and 0.533/0.753/0.740 on the second (slope +0.102, $p = 0.18$), both null, but with opposite-signed point estimates and substantially different levels. The first block alone would license the claim that the damage is a roughly constant fraction of baseline error; the second withdraws it. Reporting the growing absolute number without this would overstate a mechanism we cannot see.

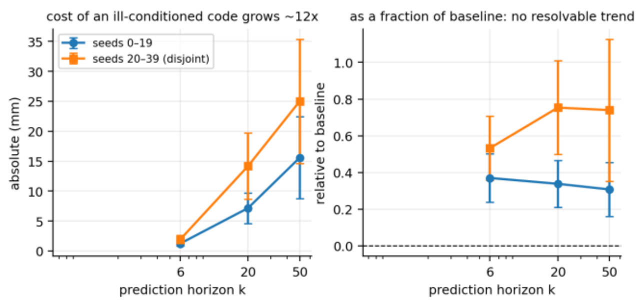

Fig. 4. How the cost of an ill-conditioned code scales with the prediction horizon, on two disjoint seed blocks. Absolute cost grows about thirteenfold; expressed as a fraction of baseline no trend is resolvable in either block, and the two blocks' point estimates carry opposite signs.

A pre-registered entry criterion confines the results above to arm2. A criterion on ten fresh seeds per body, setting no millimetre threshold and asking only whether the instrument resolves a conditioning effect, passes arm2 (+1.359 mm, $p = 0.023$) and fails arm3 (−0.007 mm, $p = 0.99$), so arm3 never entered the test and none of its numbers are invoked in support of a claim; the one quoted below is quoted against us. The criterion is defined at $k = 6$ only, so its null bounds what we can resolve there and not whether the effect exists: on the same ten seeds the conditioning cost at $k = 50$ is +32.9 mm ($p = 0.047$), uncorrected across twelve cells and outside those seeds' authorised use, which is enough to withhold the stronger claim but not to make a result.

4) Where the identity does not apply:

The identity covers re-encoding but not changing f itself. Two conditions would escape it. The core must move, or the fit must be under-determined. Across four bodies and ten settings the spread in held-out error across code initialisations (24 on the three planar bodies, 12 on `myoArm`) never exceeded 3% of that setting's minimum detectable effect projected to $n = 20$, which ranges from 0.221 mm to 1.336 mm across the four bodies. The largest observed ratio was 0.028. Four of the ten settings carry no pre-registration guarantee (Section III): their measurements stand, but the guarantee does not extend to them. In-sample fits from different starts agreed to one part in 10^5–10^6, **under a curvature-aware optimiser**. That scope is load-bearing.

On the same models, data and starts, Adam with a decayed learning rate gives a start-point spread of 1.47–1.83 mm at the two smallest adaptation budgets, 10 and 25 transitions, which is 6.7–7.2× each setting's own minimum detectable effect (`unconverged_frac` = 0.00, so not a convergence artifact). Our own five mechanisms adapt θ with Adam at $lr = 5 \times 10^{-2}$, 80 steps. Their codebase contains no L-BFGS, and the L-BFGS results above come from the separate follow-up codebase. So the code-fitting problem has one answer *under a curvature-aware rule*, and **we cannot claim our five mechanisms sit in the regime where the identity transfers from the function to the estimate**.

Pushing into the under-determined region was not measured at a size that settles the idea. Raising the code dimension to 64 or 128 against ten transitions does make start-point spread exceed that effect, but adaptation itself is not demonstrably working there. The adapted model beats the unadapted one on 2/6 and 3/6 seeds respectively, and across all four under-determined settings the figure spans 0–4 of 6. Those are raw per-seed win counts, the statistic Section IV-A replaces with a seed-level interval wherever a treatment contrast is at stake. We use them here only to show that the regime is degenerate. They would not carry a contrast.

A change of dependent variable reaches neither condition, though it does scope the nulls. Every result above uses open-loop prediction error, at six steps except where Section IV-C3 varies the horizon. A pre-registered test asked whether that choice is doing the work, and it is. When the same adapted codes are scored under closed-loop random-shooting MPC, under L-BFGS the spread across eight restarts is 0.001630 of the minimum detectable effect on the prediction metric (1.011 mm) and 2.1441 on the control metric (38.27 mm of MPC cost), both at the twenty seeds the round ran (Fig. 5), an amplification of 1316× (arm2, 20 seeds × 3 held-out bodies × 8 restarts). It is a ratio of two MDE-normalised spreads, each estimated on those twenty seeds, and we attach no interval to it. It is an order-of-magnitude statement about which metric can resolve a start-point difference, not an effect size. It holds even under the pre-registered check guarding against MPC's own sampling noise (35.8 mm against an 82.1 mm effect).

One of the five pre-registered checks was never tested at our unit of analysis, so we re-ran the round at its pre-registered design size instead of leaving it untested. `DESIGN_TASK`'s check 1 required adapted control to beat both the unadapted code and the random-action floor significantly. The four-seed run scored it on point means only and no test was ever computed, and at the seed level its contrasts were −34.0 mm ($p = 0.16$) and −91.2 mm ($p = 0.18$). At twenty seeds the check still cannot be scored, for a reason that is itself the finding. The adapted code is evaluated from eight restarts and the unadapted code from one, so the two are not comparable statistics. Across the twenty seeds the adapted arm's best restart costs 104.1 mm and its worst 188.9 mm, bracketing the unadapted code's 143.3 mm: on best-of-restarts adaptation wins by −39.2 mm (CI [−50.2, −28.2], $p = 5 \times 10^{-7}$) and on worst-of-restarts it loses by +45.6 mm (CI [+32.0, +59.2], $p = 1 \times 10^{-6}$). **We therefore withdraw check 1 rather than score it**; scoring it would need a restarted control, which this round did not run. That the answer turns on which restart is chosen is the same phenomenon the 1316× amplification measures. The twenty seeds are 0–19 and therefore include the original four, so this is the design-size completion of that round rather than an independent replication.

Two other MPC rounds resolve no adaptation benefit on their pre-registered mean-over-restarts metric. A best-of-restarts comparison is available for the second of them and for this round, but it is confounded by the restart count in the same way, so we read no direction from it (S2). The round therefore establishes one thing at the twenty seeds it ran (Fig. 5): on a control objective the choice of code start has consequences the prediction metric cannot see. It establishes nothing about whether adaptation itself pays there. We still do not claim the two conditions are jointly satisfied.

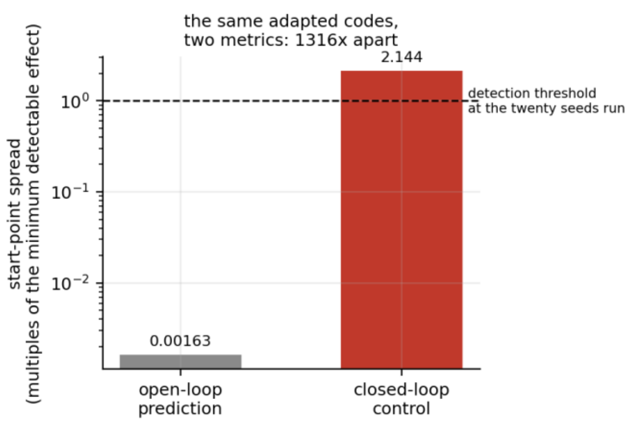

Fig. 5. The same adapted codes scored two ways. Start-point spread is 0.001630 of a minimum detectable effect under open-loop prediction and 2.1441 under closed-loop control, a 1316-fold amplification. Both are normalised by each metric's own minimum detectable effect at the twenty seeds the round ran, so the control bar's crossing of the threshold is a measurement rather than a projection.

The change of dependent variable scopes our negative results; it does not overturn them. The identity is untouched. A function left unchanged returns identical values of *any* downstream measure under an equivariant rule, control objectives included. Only the empirical half narrows. The identity does not. All our searches for a regime in which organising the code pays used the prediction metric. The calibrated one bounds any net benefit from whitening on arm2 at 0.099 mm; none of them establishes that no benefit exists on any metric. Whether one exists under a control objective is a question this round reopens without answering. This round used only the untreated arm, by design, so it shows the control metric *can* resolve start-point differences and says nothing about whether organising the code improves them. Our pre-registration names the experiment that would settle it and records that it may not be skipped. **Two attempts failed their own pre-registered checks:** the first was underpowered and set no entry condition, and the second passed the entry condition it did set and then failed downstream on three counts, one of them that adaptation did not beat the unadapted code on the pre-registered mean-over-restarts metric on a fresh seed block (S2). We add one disclosure the pre-registration did not require. The second round's achieved minimum detectable effect on the test block, computed at the seeds that round actually ran, was 9.05 mm against the 8 mm entry criterion, so applied to the realised run instead of the projection, the round would not have opened. Supplementary S2 gives both attempts in full. The gap therefore remains open, and the correct reading is that this instrument, budget-10 code adaptation scored by random-shooting MPC, cannot answer the question, not that the answer is negative.

D. *Control metrics and compute*

Neither control metric supports the joint-over-staged ranking once analysed at the seed level. Both metrics are exploratory (Section III). Random-shooting MPC makes staged 27.7 mm worse, with a seed-level interval of [−33.7, +89.2] ($n = 4$, $p = 0.25$; one of four seeds reverses). Learned-policy SAC+Dyna gives joint − staged return-AUC +0.10, [−0.06, +0.26] ($n = 3$, $p = 0.11$). The world model does beat model-free SAC (+0.99, [+0.47, +1.51], $p = 0.015$), but only at $n = 3$ seeds, at or below the sizes at which we declined to interpret the two contrasts above, so we report it and do not lean on it. The staging conclusion therefore rests on the prediction proxy alone, and we can no longer claim it is corroborated outside it.

The whole pipeline runs in minutes of CPU on a single consumer laptop in pure MuJoCo, with the pilot screen completing in 43 s. Every experiment above is re-runnable at that cost.

V. DISCUSSION

The part of this work that generalises beyond our arms is not a measurement. When a shared dynamics core is frozen and only a low-dimensional code is adapted under a rule equivariant to the map, re-coordinatising that code by an invertible matrix is inert by construction, and orthogonalising it, rescaling it and whitening it are all excluded before an experiment is run. What makes that useful rather than trivial is that the conditions the argument needs are the ones a practitioner is most likely to violate without noticing: nonlinear re-encodings fall outside the family, and disentangling generally asks for one; and the identity carries from the function to the *estimate* only under an equivariant adaptation rule, which Adam is not.

Those two conditions leave two routes by which reorganising a code might still pay, and we could put a calibrated bound on one. Against a positive control demonstrated on the same seeds to resolve a +1.234 mm conditioning effect, whitening the code is worth +0.025 mm on net, and its per-cell displacement is symmetric where the control's is systematic (Section IV-C2). Both figures are arm2, and the one attempt to show the same effect on a second body did not resolve it (Section IV-C3). That is a bounded statement rather than an absence of evidence, which is why a reader need not take the null on trust.

The other route, nonlinearity, we measured but did not close. The gated map is what we would put to anyone reading reorganisation as disentanglement: built to be the identity at every training code, it preserves leave-one-body-out decodability to machine precision by construction and still costs +0.875 mm. Keeping a code exactly as readable does not keep it equally useful, and a proxy that many representation-learning arguments rest on comes apart, on our metric, from the property it stands in for.

The empirical half belongs to an older tradition than the algebra. Our one directional result sits with the staging tradition Section II sets out: Elman [25], its failed replication by Rohde and Plaut [26], and Bongard's [27] move from input to body. All specify an ordering of experience, and so does ours. There is no competence gate anywhere in our code, so the result speaks to that tradition and not to competence-triggered development. Read against Rohde and Plaut in particular, a staging cost is not an anomaly.

That lineage also fixes what the null does not license. What fails is the *engineering* hypothesis, and this is **not evidence for or against RR as an account of human development**: a null on a simulated arm cannot bear on such a theory, and RR proper is endogenous whereas our steps are experimenter-driven. The point is sharpest on Karmiloff-Smith's E1 level, at which knowledge is redescribed into a format available *as data to the system itself*. Our decodability shows only that morphology information is linearly recoverable by an external probe; the available-to-the-agent criterion is **not met**, and external decodability is strictly below it. Nor did the theory promise what the engineering hope wants of it: RR operates beyond behavioural success and is paid in accessibility, not accuracy, so a transfer metric that holds the task fixed and changes the body is where it predicts no benefit.

A deflationary reading of the positive half is that θ is a disentangled system-identification code of the same character as RMA's extrinsics latent. We accept it; discovering such a code is not our contribution, and what the developmental framing adds is the staging test and the characterisation of which factors the code isolates. One bridge is speculative and we mark its limit: a morphology code developed by a forward model might be read as the *dynamics* analogue of a body schema, but the analogy is by functional role only. The code shares no representational content with a postural body schema and **should not be cited as an emergent body schema**. The world model is body-only by construction, so none of this is a claim about self/world separation.

If reorganisation does not pay here, it is worth asking where it does. Fine-tuning of large pre-trained models sits in the under-determined regime, and there a low-rank constraint on the adaptation space matches or beats full fine-tuning [37]. That structure is a constraint on the adaptation space, not a better encoding of an existing code, and our body codes are already at that endpoint: a four-dimensional θ against a frozen core *is* such a subspace. Starting at the destination is the most economical account of why the journey buys nothing here, and it suggests a recommendation we did not test: seek structure as a constraint on an over-parameterised adaptation space, not as a reorganisation of a code that is already minimal.

The same reasoning suggests where a payoff would be visible if one exists. If redescription buys accessibility, it should appear where recombination is *required*: novel goals composed from mastered parts, reuse of a body model across tasks it was not fit for, any setting where the code must be manipulated rather than re-estimated. We did not test that, and this is a post-hoc reframing of a pre-registered null rather than a result. We record it because the alternative reading, that our null counts against RR, is unsupported. For anyone building such systems the practical residue is short: expose the world model to a distribution of bodies, and do not expect a fixed reordering of that experience to pay on a prediction metric.

VI. LIMITATIONS

Bodies. The main experiments use 2-, 3- and 4-link planar arms; 3D morphologies are untested. Re-running mechanism 5 unchanged on `myoArm` (20 joints, 34 muscles) returns a null (joint − staged = +0.19 mm, seed-level CI [−24.08, +24.47]) whose minimum detectable effect at the 4 seeds × 6 held-out bodies actually run is 22.4 mm on the project's design-size constant (S1), against the 0.90 mm effect measured on arm2, so it is uninformative rather than disconfirming. The shortfall is structural, not budgetary: what remains after the arms share initialisation and training data is the divergence of two optimisation trajectories, 84% of the between-seed variance, and pairing cannot remove it because the trajectories diverge by seeing different data in a different order, which is the treatment itself. Adding bodies barely helps: 6 → 24 bodies moves the MDE only from 22.4 mm to 21.0 mm at four seeds, and 50 seeds × 24 bodies would still leave 5.9 mm. Neither was run; reaching 0.90 mm would need of order two thousand seeds or a lower-variance training procedure. The decodability result does extend to that body (all three factors significant on 4/4 seeds), so the limitation is specific to the staging result. Supplementary S1 gives the decomposition.

Reproducibility of the mechanism-5 figures. The adaptation start point is drawn from a global RNG and not from the reported seed, so the absolute AUCs of mechanism 5 are not reproducible from the published seed, and neither is the paired contrast at the precision we print. Restoring the call order moves the gap from −0.90 mm to −1.282 mm at the same n = 5 and −1.374 mm at n = 20, a 42–53% change. That −1.374 mm is the *unmatched* contrast the original run made; the step-controlled contrasts we lead with (Section IV-A) come from the same round. Direction and significance survive and the re-measured effect is larger (seed-level CI at n = 20 [−1.843, −0.905]), so no conclusion turns on it, but −0.90 mm should be read as reproducible only to within about a factor of 1.5.

Task family and dependent variable. All bodies are single-arm reaching systems and the primary metric is open-loop prediction error. The two control metrics do not corroborate the staging ranking at seed level (Section IV-D), and a pre-registered test shows the choice of metric is not neutral. Start-point differences worth 0.001630 of a minimum detectable effect under prediction are worth 2.1441 under closed-loop control, both at twenty seeds (Section IV-C4). The negative results are therefore scoped to the prediction metric. The experiment that would settle whether organising the code pays under a control objective is named in our pre-registration as a step that may not be skipped. Both attempts failed their own pre-registered checks (Supplementary S2), and this is the most consequential piece of unfinished work here.

The residuals-per-unknown regularity. This regularity is not a law, and we record it as an observation that appears nowhere in Section IV. Across 14 settings, the influence of start-point choice tracks the number of residuals in the code fit per free code parameter (Spearman −0.923, $p = 2.6 \times 10^{-6}$), but four settings failed the project's own manipulation check; restricted to the ten interpretable ones the correlation weakens to −0.811 ($p = 4.4 \times 10^{-3}$). Of those ten, a further four are the settings whose pre-registration guarantee Section III withdraws, so the same caveat applies here. A pre-registered out-of-sample test on `myoHand`, with predictions and a ±2 SD band fixed before the run, placed three of four new settings outside the band.

Adaptation optimiser. All five mechanisms adapt θ with Adam ($lr = 5 \times 10^{-2}$, 80 steps); their codebase contains no L-BFGS, and the L-BFGS measurements of Section IV-C come from the separate follow-up codebase. The identity transfers from the function to the estimate only for maps the adaptation rule is equivariant under, and Adam's class is the permutations alone (Section IV-C). The start-point stability that closes the under-determination escape is likewise an L-BFGS result. Under Adam the same spreads are 6.7–7.2× their own settings' minimum detectable effects rather than under 3% of them. We therefore cannot claim these mechanisms sit in the regime where the identity binds their estimates, only that it binds the function they compute.

Non-claims. We claim no mastery trigger, no monotone relation between mass decodability and mechanical coupling, no general ordering of joint over staged decodability, and no control-metric corroboration of the staging result; none is supported at the level Section III fixes. Four results survive, each resting on more than a pilot: the invariance argument, which is algebra whose magnitude is measured on one twenty-seed round; the staging-worse result at 20 seeds on two bodies, under a manipulation-check criterion substituted post hoc (Section IV-A); the factor-specific decodability characterisation at 12 seeds, which is exploratory (Section III); and the calibrated conditioning null on arm2.

VII. CONCLUSION

Training a body-dynamics world model across a distribution of bodies yields a low-dimensional unsupervised code whose alignment to true morphology we characterise under leave-one-body-out cross-validation, and whose decodability is *factor-specific*: damping and gain travel across every body we measured, while mass is unstable across seeds rather than absent. Such a code is expected from context-conditioned dynamics learning; our modest positive is that an unsupervised one nonetheless decodes named physical scalars.

The developmental lever one might expect, master-then-differentiate redescription or staging, is not what produces or exploits it. Neither of the two mechanisms carrying reportable data shows the hoped-for benefit: the redescription one is a null and the staging one a reversal, with staging *worse* than joint training at 20 seeds on two of three bodies and the joint arm's step count cut below the staged arm's (Section IV-A, S8).

Two things make that stronger than a bare null. A large sub-family of the interventions the hope reaches for is excluded by an identity rather than by measurement: with the core frozen, re-encoding the code cannot change held-out error under an equivariant adaptation rule. And the second of the identity's two loopholes, conditioning under a non-equivariant optimiser, was tested against a positive control that worked. Whitening the code moves held-out error by +0.025 mm on net ($p = 0.68$), the wrong sign, whereas degrading its conditioning costs +1.234 mm ($p = 2.2 \times 10^{-6}$). Section IV-C3 reports a round whose own control failed on a second body, which is why we keep the arm2 result and the arm3 non-result separate.

This does not license "redescription fails": the mechanism Karmiloff-Smith describes is not paid in the currency our transfer metric measures (Section V). It licenses something narrower and more useful to anyone building such systems. The *engineering* hope that attaches a performance currency to that mechanism is not borne out on our proxy, and what the code does carry, linear decodability by an external probe, arrives in a model that has simply experienced body variation. That all of this runs in minutes on a single CPU lowers the barrier to studying such questions further.

DATA, CODE AND PRE-REGISTRATION AVAILABILITY

Simulator, training code, analysis scripts, every pre-registration document, and the stored result files behind every number in this paper are available at [repository URL]. Each pre-registration fixes the decision rule and the manipulation checks in advance; twenty-one of the twenty-four also contain the conclusion paragraph for every outcome written verbatim. `PREREG_HASHES.md` archives design-document fingerprints and, for those that existed when it was opened, creation times; `README_PREREGISTRATIONS.md` indexes every round in English, since the design documents themselves are in Chinese. Where that record does not reach (the settings named in Section III, and sixteen result files from the exploratory rounds), the repository's `CORRECTIONS.md` gives the itemised account.

DECLARATION OF GENERATIVE AI USE

Generative AI (Claude, Anthropic) was used to assist with implementation of experiment and analysis code, and with drafting and editing the manuscript text. It was not used to generate data, results, or references. All experimental designs, pre-registered decision rules and manipulation checks, statistical choices, and the interpretation of every result were determined by the author, who takes full responsibility for the content. Every numerical claim in this paper was re-derived from the stored raw result files by verification scripts included in the repository.

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